

1-v-ALL RDA & 1D Dilated CNN Control System for Hand Gesture Recognition Using Two EMG Channels

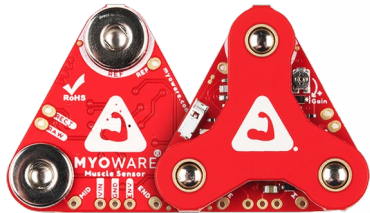
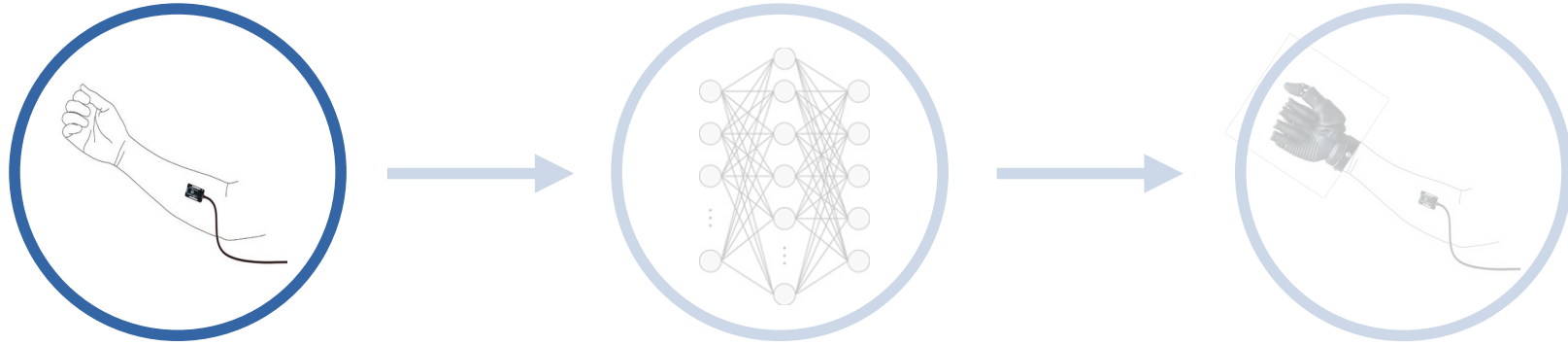
Barret Feusi

Problem

Most smart hand prostheses with gesture classification capabilities require high quality data from a high density sensor setup to reliably distinguish between different hand gestures for proper classification, which can raise the projected cost of the system significantly. Additionally, algorithms that are not robust fall in performance when subjected to input data that is far noisier than original training samples.

Ownership of these devices requires efficient control.

Primary Objective



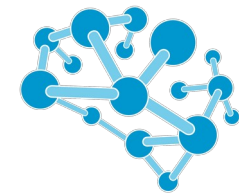
EMG



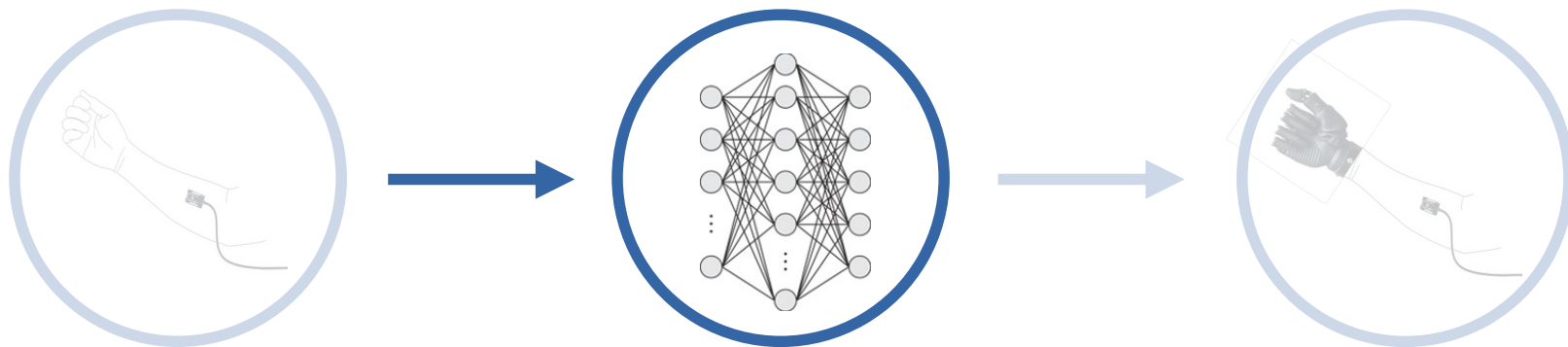
ESP32-S3

To make a real-time biosignal acquisition and processing module...

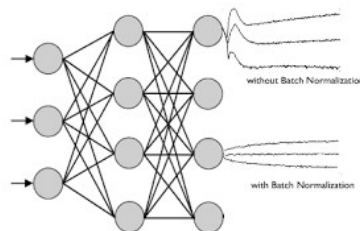
Primary Objective



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...with robust filtering

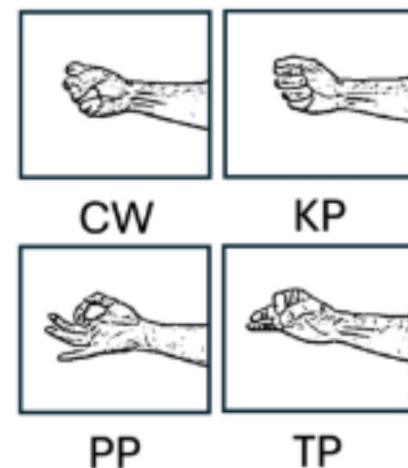


and normalization...

Primary Objective



...to predict one of four hand gestures
for controlling an **affordable**
smart hand prosthesis.



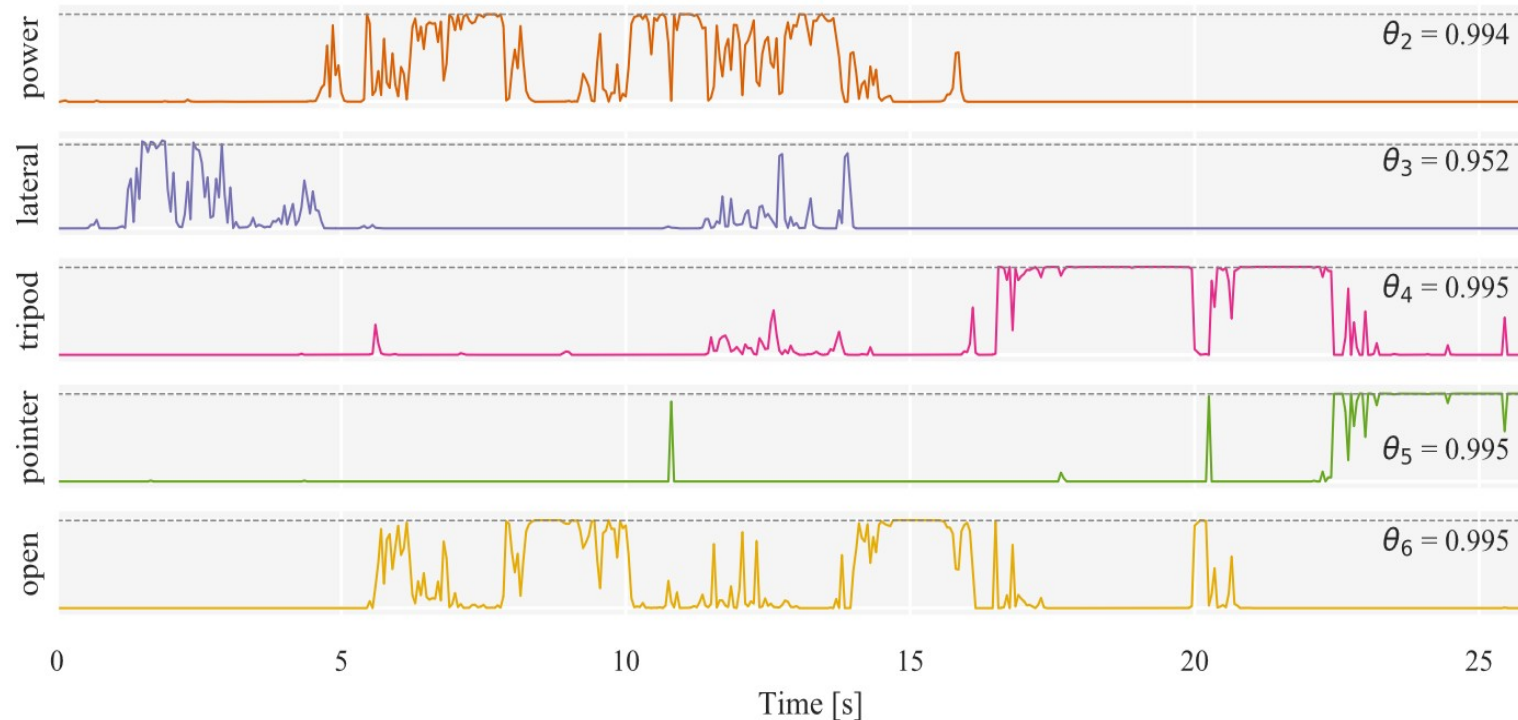
Background



Making a robust classification algorithm for a smart prosthetic can be done by aggressively **minimizing false positives**, while **utilizing the most effective feature sets** in the input data to properly distinguish between the different intended gestures, thereby optimizing the number of true positive predictions it can act upon.

Research

Posterior class probabilities



“Machine Learning-Based sEMG Signal Classification for Hand Gesture Recognition” [3]

Confidence rejection thresholds for 1-v-All RDA models can result in a “gatekeeper” that will not accept a prediction unless the posterior probability exceeds the threshold optimized for highest F1 score.

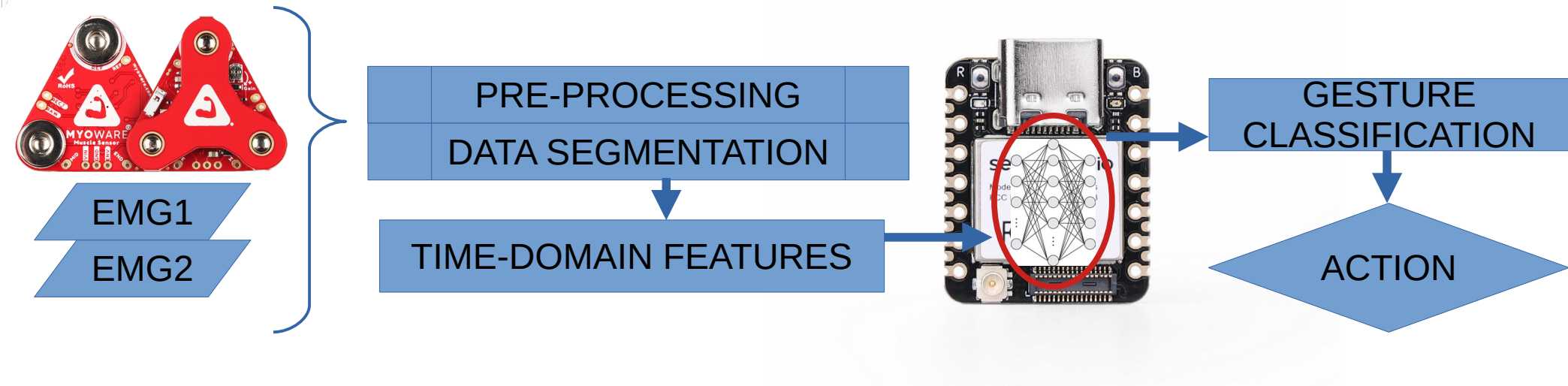
Research

When trained on time-domain and temporal-spatial features (features dealing with wave energy and sparsity), 1D Dilated CNN models offer some of the best accuracy on various EMG datasets, like GrabMyo and FORS-EMG.

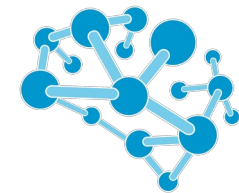
TABLE III
GESTURE RECOGNITION FROM TIME-DOMAIN DESCRIPTORS USING
TRADITIONAL MACHINE LEARNING AND DEEP LEARNING MODELS
TESTED ON THE GRABMYO DATASET.

Models	ACC	P	R	F1
LDA	84.26	0.85	0.84	0.84
SVM	93.12	0.93	0.93	0.93
KNN	93.86	0.94	0.94	0.94
Random Forest	93.6	0.94	0.94	0.94
Voting Ensemble	94.29	0.95	0.94	0.94
Bagging KNN	94.33	0.94	0.94	0.94
Bagging SVM	93.38	0.94	0.93	0.93
Adaboost	93.61	0.94	0.94	0.94
1D Dilated CNN	97.00	0.97	0.97	0.97
1D Dilated CNN-LSTM	96.53	0.97	0.97	0.97

Proposed Solution

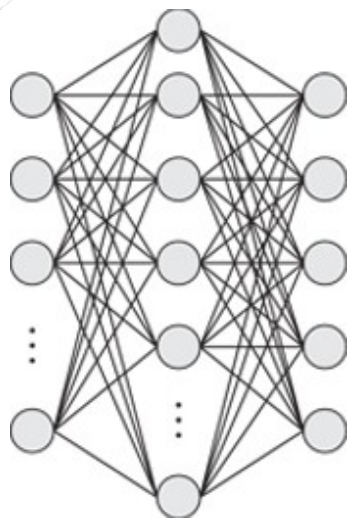


Proposed Solution



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4 1-v-All RDA + 1D Dilated CNN ML Model

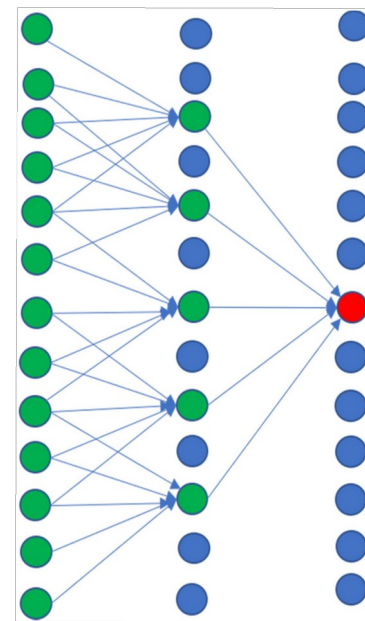


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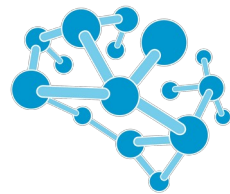
“Gatekeeper” Model

+



Second Opinion
Model

Development Process

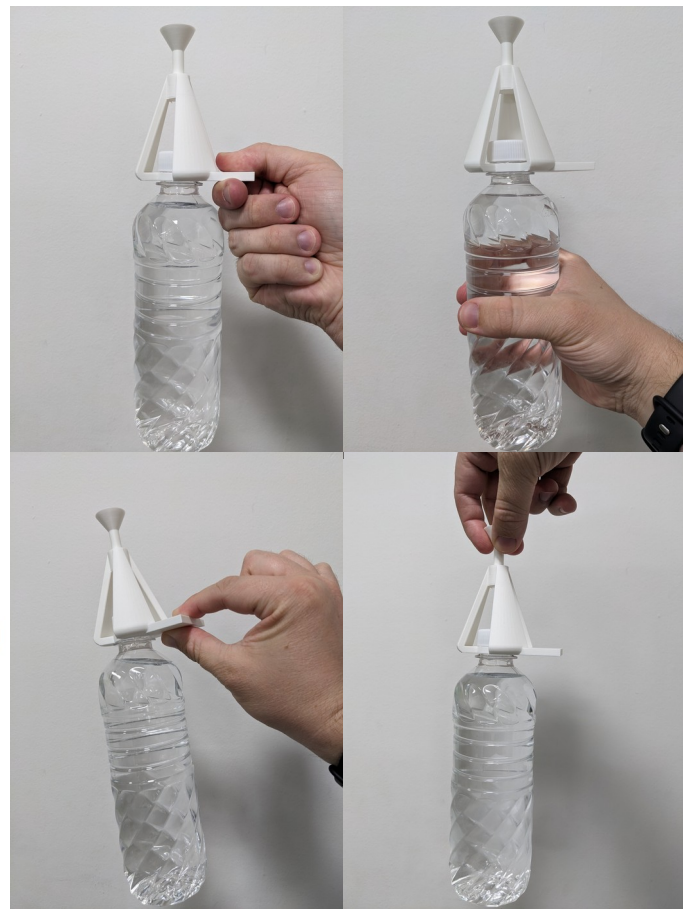
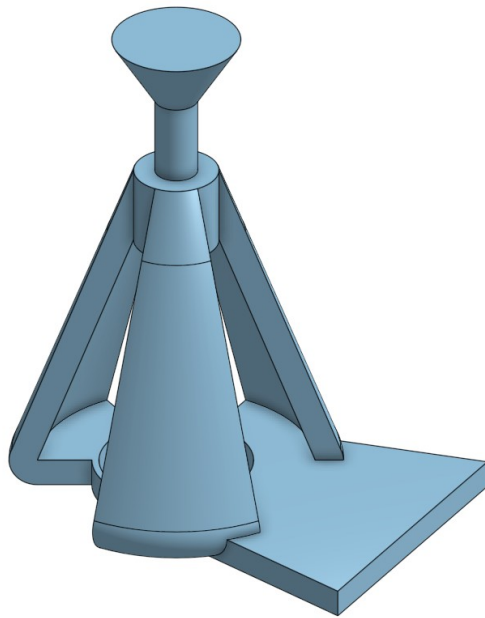


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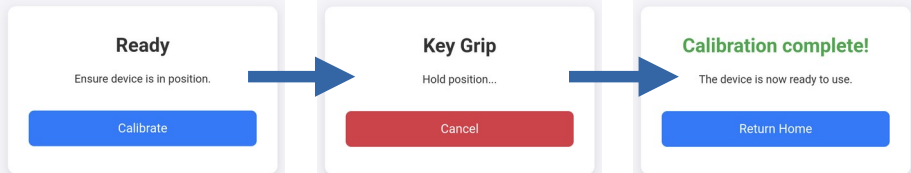
ML Model Research



Testing Design



Data Collection



Development Process

ML Development

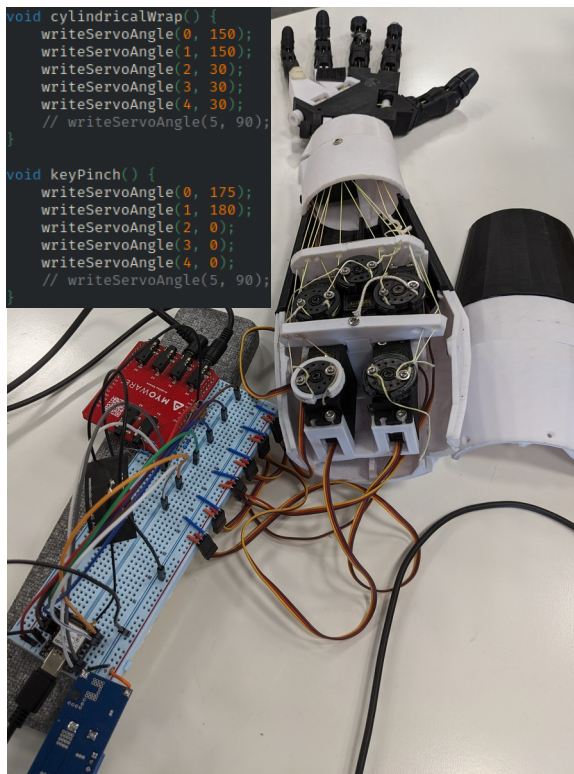
Pre-Trained Models



TFLite Models



C++ Script for ESP32



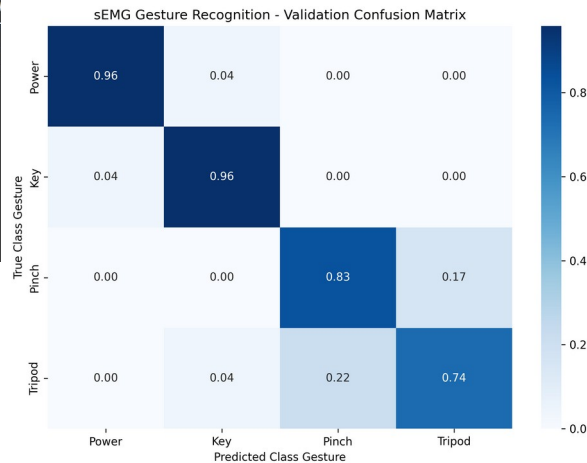
Gesture Control & Wiring

Validation Testing

```
Calculating confidence-based rejection thresholds...  
Acceptance Rate: 0.3364  
Accuracy on Accepted Predictions: 1.0000  
Class 0 Threshold: 0.7952  
Class 1 Threshold: 0.8057  
Class 2 Threshold: 0.8141  
Class 3 Threshold: 0.7834
```

Average Conditional Accuracy across 10 folds: 1.0000

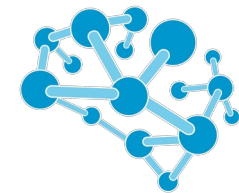
```
--- Final Model Show  
Actual | Predic  
-----  
Pinch  | Tripod  
Key    | Key  
Pinch  | N/A  
Power  | Power  
Power  | N/A  
Pinch  | Pinch  
Pinch  | N/A  
Tripod | N/A  
Pinch  | N/A  
Pinch  | N/A
```



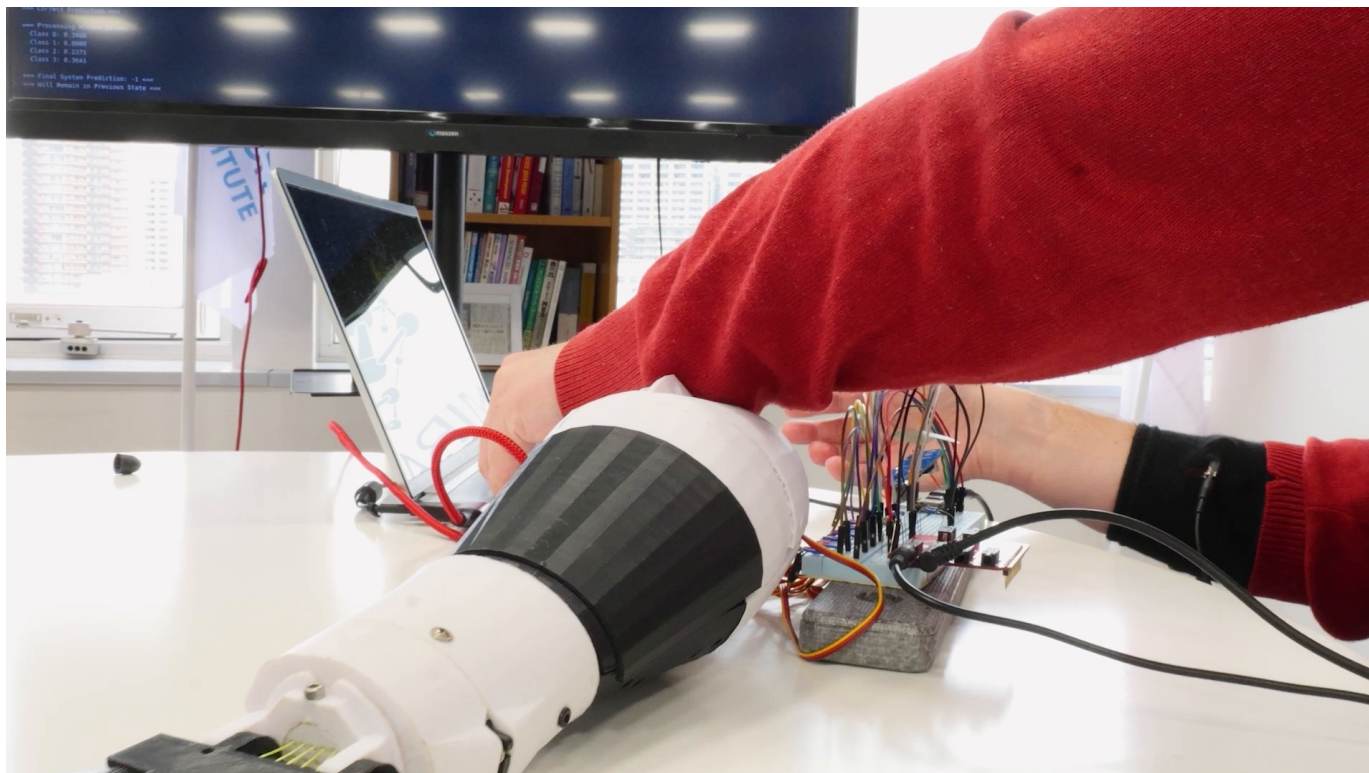
Prototype



Prototype



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Results & Evaluation

Average Accuracy – Accepted Predictions:
98%

Calculating confidence-based rejection thresholds...

Acceptance Rate: 0.3364

Accuracy on Accepted Predictions: 1.0000

Class 0 Threshold: 0.7952

Class 1 Threshold: 0.8057

Class 2 Threshold: 0.8141

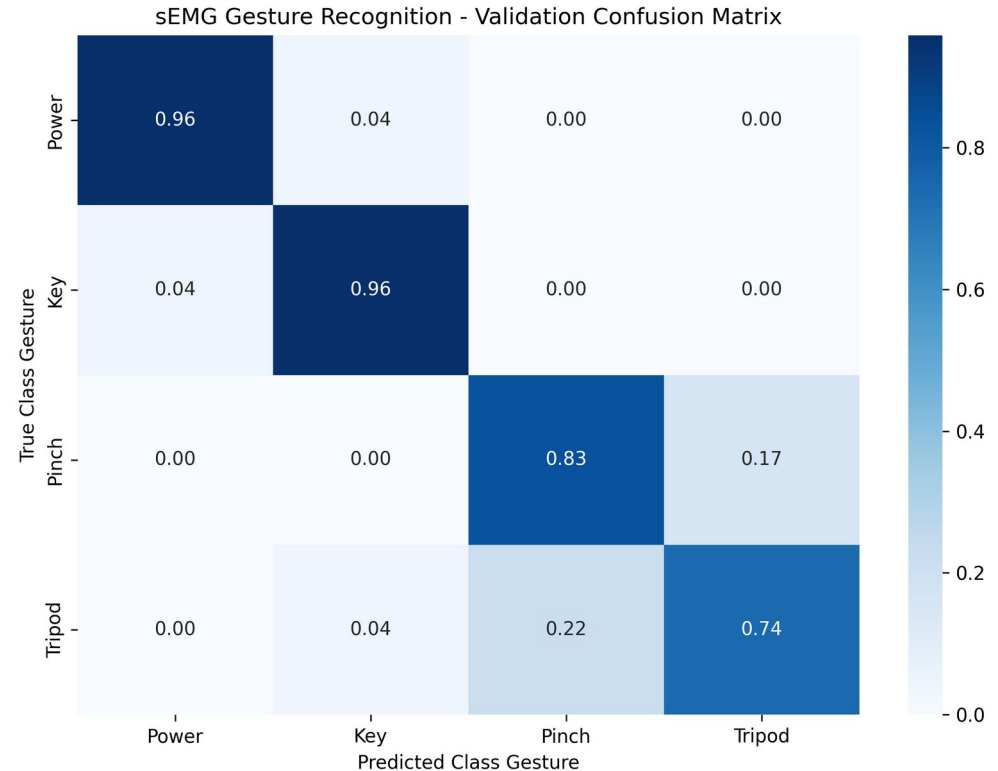
Class 3 Threshold: 0.7834

Average Conditional Accuracy across 10 folds: 1.0000

--- Final Model Showcase (10 Random Samples from Fold 10 Validation Set)

Actual	Predicted	Prediction	Confidence	Status
Pinch	Tripod	Tripod	0.8005	No
Key	Key	Key	0.9466	Yes
Pinch	N/A	Pinch	0.6530	N/A
Power	Power	Power	0.8390	Yes
Power	N/A	Power	0.7463	N/A
Pinch	Pinch	Pinch	0.8370	Yes
Pinch	N/A	Tripod	0.5895	N/A
Tripod	N/A	Tripod	0.7309	N/A
Pinch	N/A	Pinch	0.7353	N/A
Pinch	N/A	Pinch	0.6307	N/A

Average Accuracy – Validation Set:
87%



Results & Evaluation



In a test of **32** novel pre-acquired labeled data segments that were used for validation of the model, the combined model operated on predictions for **14** of the segments and got all **14** of the predictions correct.

However, at a rate of data collection and operating speed of about 50 ms per operation, this means that the time needed to safely make an accurate prediction is around **114.2 ms**.

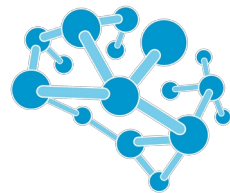
Innovation



This current design proves that it can:

- 1) Eliminate false positives meaningfully and only act on predictions when certain enough or in agreement with a secondary model.
- 2) Fit all processing and five machine learning pre-trained models on a single ~\$5 USD microcontroller.
- 3) Leverage the aspects that make 1-v-All RDA models robust and 1D Dilated CNN models performant to strike a balance between being robust and performant.

Impact & Future Work



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This combined model has shown that it is capable of operating on low-dimensional biosignal data while working on an affordable microprocessor, thus dramatically reducing the projected cost of a smart prosthetic that utilizes this model or a similarly robust and performant algorithm.

I will continue my work on this research to experiment more with confidence and consensus between models and to prove that models like these can be robust even when subjected to challenging situations that affect the data, like sweat.

Conclusion

In light of the latest technological and scientific advancements, especially in complex algorithms capable of discovering patterns and key features in large data sets, the market for more capable, affordable smart prostheses is coming to fruition.



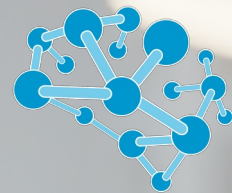
References



- 1) Jaramillo-Yáñez, Andrés, Marco E. Benalcázar, and Elisa Mena-Maldonado. 2020. "Real-Time Hand Gesture Recognition Using Surface Electromyography and Machine Learning: A Systematic Literature Review." *Sensors* 20(9):2467. doi:[10.3390/s20092467](https://doi.org/10.3390/s20092467).
- 2) Young, Peyton R., Kihun Hong, Eden J. Winslow, Giancarlo K. Sagastume, Marcus A. Battraw, Richard S. Whittle, and Jonathon S. Schofield. 2025. "The Effects of Limb Position and Grasped Load on Hand Gesture Classification Using Electromyography, Force Myography, and Their Combination." *PLOS ONE* 20(4):e0321319. doi:[10.1371/journal.pone.0321319](https://doi.org/10.1371/journal.pone.0321319).
- 3) Krasoulis, A., Vijayakumar, S., & Nazarpour, K. 2020. "Multi-Grip Classification-Based Prosthesis Control With Two EMG-IMU Sensors." *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(2), 508–518. doi:[10.1109/TNSRE.2019.2959243](https://doi.org/10.1109/TNSRE.2019.2959243)
- 4) Aarotale, Parshuram N., and Ajita Rattani. 2024. "Machine Learning-Based sEMG Signal Classification for Hand Gesture Recognition." doi: [10.48550/arXiv.2411.15655](https://doi.org/10.48550/arXiv.2411.15655).

* First and last images generated with Gemini AI, all other images made or shot by author.

Q & A



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Thank you!

